

Final Project - Data 200

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Loading and Inspecting Project 1 Dataset

- This section imports the required Python libraries and loads the Project 1 dataset, `StudentPerformanceFactors.csv`, into a pandas DataFrame. The file path is constructed using the `os` module to ensure the dataset is correctly located in the working directory. After loading the data, I inspect the distribution of the `School_Type` variable to understand how observations are split between public and private schools. This step provides an initial overview of the dataset structure and confirms that school type is a meaningful categorical variable for subsequent analysis.

```
In [36]: import os
import pandas as pd
import numpy as np
import requests
from io import StringIO

In [37]: data = 'StudentPerformanceFactors.csv'
csv_path = os.path.join(os.getcwd(), data)
print(csv_path)

df = pd.read_csv(data)

Out[38]: School_Type
Public      4598
Private    2009
Name: count, dtype: int64

In [38]: df['School_Type'].value_counts()
```

Web scrape/ API function

- This project uses the OECD SDMX REST API to retrieve data on average class size by type of educational institution. The API endpoint is structured around a dataflow identifier that specifies the education dataset, followed by dimension codes that define education level, institution type, frequency, and time period. These dimensions are encoded directly in the URL path and query parameters, allowing targeted data extraction.
- The function exploits this structure by dynamically inserting user-defined start and end years into the API query. This allows the function to retrieve average class size data for different time ranges, which substantially changes the observations returned by the API. The function returns the data in CSV format with labeled dimensions, making it straightforward to parse into a pandas DataFrame.
- The dataset contains observations across multiple countries, education levels, and institution types (public and private). This raw data can then be filtered and aggregated to produce tidy summaries suitable for comparison with student-level performance data. The function enables flexible and reproducible access to OECD education statistics for further analysis.

```
In [39]: def get_oecd_avg_class_size(start_year, end_year):
    """
    Pull OECD average class size data by school type.
    Customizable by start and end year.
    """
    url1 = (
        "https://sdmx.oecd.org/public/rest/data/"
        "?dc=EDU.IMEP.DSD_EAG_UOE_NON_FIN_PERS&FREQ=AN&INST_EDU_PUB=INST_EDU_PRIV...T."
        "&startPeriod={start_year}&endPeriod={end_year}"
        "&format=json&withLabels="
        "&dimensionAtObservation=AllDimensions"
    )
    url2 = (
        "https://sdmx.oecd.org/public/rest/data/"
        "?dc=EDU.IMEP.DSD_EAG_UOE_NON_FIN_PERS&FREQ=AN&INST_EDU_PUB=INST_EDU_PRIV...T."
        "&startPeriod={start_year}&endPeriod={end_year}"
        "&format=json&withLabels="
        "&dimensionAtObservation=AllDimensions"
    )
    response = requests.get(url2)
    df_api = pd.read_csv(StringIO(response.text))
    return df_api

In [40]: df_api = get_oecd_avg_class_size(2023, 2023)
df_api.head()
```

	STRUCTURE	STRUCTURE_ID	STRUCTURE_NAME	ACTION	REF_AREA	Reference area	EDUCATION_LEV	Education level	MEASURE	Measure	...	CONF_STATUS	Confidentiality status	COMMENT_OBS	Observation comment	DECIMALS	Decimals	TIME_PI
0	DATFLOW	OECD.EDU.IMEP.DSD_EAG_UOE_NON_FIN_PERS@DF_UOE_...	Average class size	I	AUT	Austria	ISCED11_1	Primary education	CLS	Student classes	...	NaN	NaN	NaN	NaN	2	Two	
1	DATFLOW	OECD.EDU.IMEP.DSD_EAG_UOE_NON_FIN_PERS@DF_UOE_...	Average class size	I	AUT	Austria	ISCED11_1	Primary education	CLS	Student classes	...	NaN	NaN	NaN	NaN	2	Two	
2	DATFLOW	OECD.EDU.IMEP.DSD_EAG_UOE_NON_FIN_PERS@DF_UOE_...	Average class size	I	AUT	Austria	ISCED11_2	Lower secondary education	CLS	Student classes	...	NaN	NaN	NaN	NaN	2	Two	
3	DATFLOW	OECD.EDU.IMEP.DSD_EAG_UOE_NON_FIN_PERS@DF_UOE_...	Average class size	I	AUT	Austria	ISCED11_2	Lower secondary education	CLS	Student classes	...	NaN	NaN	NaN	NaN	2	Two	
4	DATFLOW	OECD.EDU.IMEP.DSD_EAG_UOE_NON_FIN_PERS@DF_UOE_...	Average class size	I	DEU	Germany	ISCED11_1	Primary education	CLS	Student classes	...	NaN	NaN	NaN	NaN	2	Two	

5 rows × 58 columns

```
In [41]: print(df_api.columns.tolist())

needed_cols = ['Type of educational institution', 'Education level', 'Reference area', 'MEASURE', 'OBS_VALUE', 'TIME_PERIOD']

['STRUCTURE', 'STRUCTURE_ID', 'STRUCTURE_NAME', 'ACTION', 'REF_AREA', 'Reference area', 'EDUCATION_LEV', 'Education level', 'MEASURE', 'Measure', 'EDUCATION_TYPE', 'Education type', 'INTENSITY', 'Intensity', 'EDUCATION_FIELD', 'Field of education', 'ORIGIN', 'Origin', 'FREQ', 'Frequency of observation', 'ORIGIN', 'Origin', 'DESTINATION', 'INSTITUTION', 'Type of educational institution', 'MOBILITY', 'Unit of measure', 'SEX', 'Sex', 'AGE', 'Age', 'TIME_PERIOD', 'Time period', 'OBS_VALUE', 'Observation value', 'REF_YEAR_AGE', 'Reference year start', 'REF_YEAR_END', 'Reference year end', 'OBS_STATUS', 'Observation status', 'CONF_STATUS', 'Confidentiality status', 'COMMENT_OBS', 'Observation comment', 'DECIMALS', 'Decimals', 'TIME_PERIOD_COLLECT', 'Time period collection', 'UNIT_MULT', 'Unit multiplier']

In [42]: df_small = df_api[needed_cols].copy()
df_small.head(10)
```

	Type of educational institution	Education level	Reference area	MEASURE	OBS_VALUE	TIME_PERIOD
0	Independent private educational institutions	Primary education	Austria	CLS	NaN	2023
1	Government dependent private educational insti...	Primary education	Austria	CLS	NaN	2023
2	Government dependent private educational insti...	Lower secondary education	Austria	CLS	NaN	2023
3	Independent private educational institutions	Lower secondary education	Austria	CLS	NaN	2023
4	Government dependent private educational insti...	Primary education	Germany	CLS	NaN	2023
5	Independent private educational institutions	Primary education	Germany	CLS	NaN	2023
6	Government dependent private educational insti...	Lower secondary education	Germany	CLS	NaN	2023
7	Independent private educational institutions	Lower secondary education	Germany	CLS	NaN	2023
8	Independent private educational institutions	Primary education	Australia	CLS	NaN	2023
9	Independent private educational institutions	Lower secondary education	Australia	CLS	NaN	2023

Conversion to Tidy Data

- The raw OECD API dataset contains multiple observations across countries, education levels, years, and types of educational institutions. To convert this dataset into a tidy format suitable for analysis, I first filtered the data to retain only observations where the measure corresponds to `Average class size` (CLS) and where the institution type is either public or private educational institutions. I then removed observations with missing values in the class size variable to ensure data quality.
- After filtering, I aggregated the data by type of educational institution and calculated the mean of the observed class size values. This aggregation collapses the dataset into one row per institution type, which aligns with the tidy data principle that each variable forms a column and each observation forms a row. Finally, I renamed variables for clarity and created a simplified `School_Type` variable to distinguish between public and private institutions. The resulting `tidy_data` contains a clear and concise summary of average class size by school type and is saved for use in subsequent visualization and analysis.

```
In [43]: df_filt = df_small.loc[
    (df_small['MEASURE'] == 'CLS') &
    (df_small['Type of educational institution'].isin(['Public educational institutions', 'Private educational institutions']))
]

df_filt = df_filt.dropna(subset=['OBS_VALUE'])

df_filt.head(10)
```

	Type of educational institution	Education level	Reference area	MEASURE	OBS_VALUE	TIME_PERIOD
109	Public educational institutions	Lower secondary education	Australia	CLS	21.69175	2023
112	Private educational institutions	Lower secondary education	Australia	CLS	18.65023	2023
114	Public educational institutions	Lower secondary education	Austria	CLS	21.35718	2023
115	Private educational institutions	Lower secondary education	Austria	CLS	21.13613	2023
116	Private educational institutions	Lower secondary education	Chile	CLS	33.08763	2023
118	Public educational institutions	Lower secondary education	Chile	CLS	29.21266	2023
121	Public educational institutions	Lower secondary education	Colombia	CLS	29.99283	2023
122	Private educational institutions	Lower secondary education	Colombia	CLS	22.26511	2023
127	Private educational institutions	Lower secondary education	Costa Rica	CLS	19.80418	2023
128	Public educational institutions	Lower secondary education	Costa Rica	CLS	30.54614	2023

```
In [44]: tidy_data = df_filt.groupby('Type of educational institution')['OBS_VALUE'].agg('mean')
#tidy_data.rename(columns={'OBS_VALUE':'Average class size'}, inplace = True)

tidy_data = tidy_data.reset_index()

tidy_data.head()
```

	Type of educational institution	OBS_VALUE
0	Private educational institutions	20.068596
1	Public educational institutions	21.869572

```
In [45]: tidy_data=tidy_data.rename(columns={'OBS_VALUE':'Average class size'})

tidy_data['School_Type'] = tidy_data['Type of educational institution'].replace({'Public educational institutions': 'Public', 'Private educational institutions': 'Private'})

tidy_data.head()
```

	Type of educational institution	Average class size	School_Type
0	Private educational institutions	20.068596	Private
1	Public educational institutions	21.869572	Public

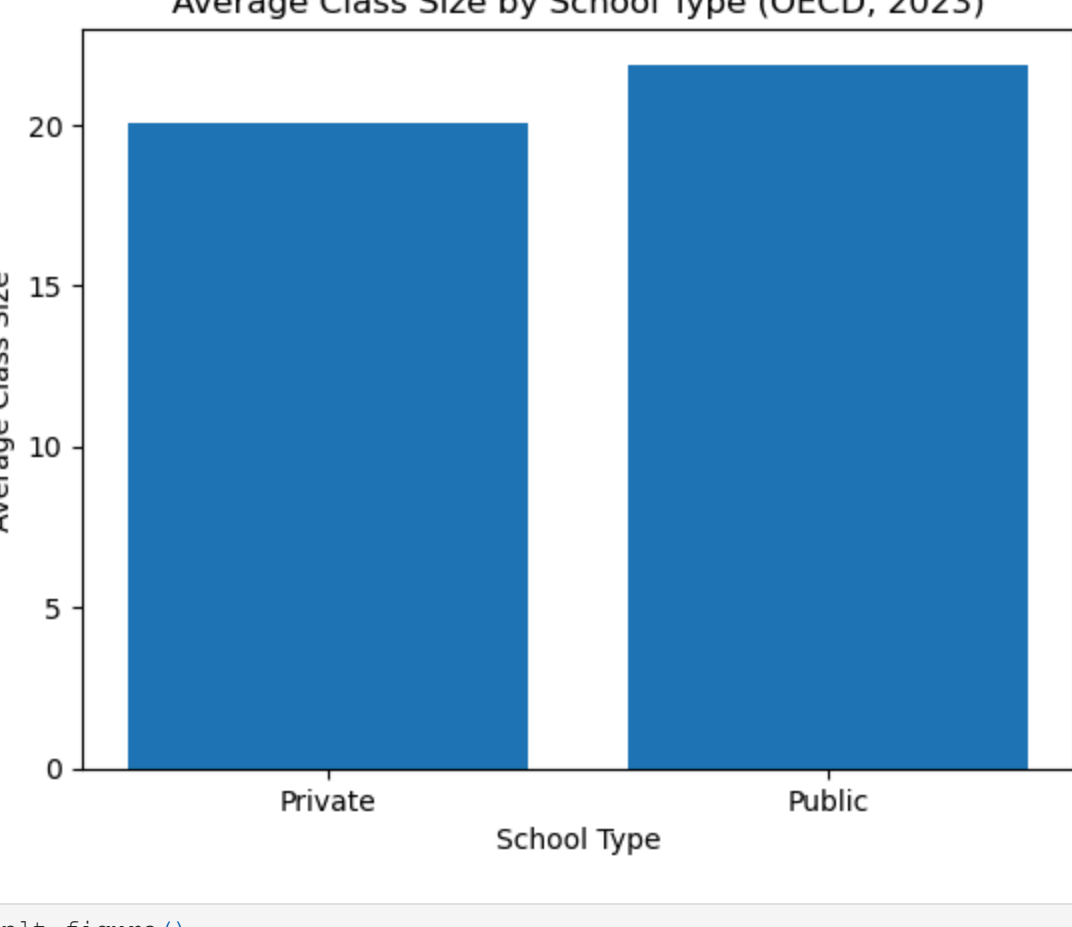
```
In [46]: tidy_data.to_csv('oecd_class_size_by_school_type_api.csv')
```

Extracted Data Visualization

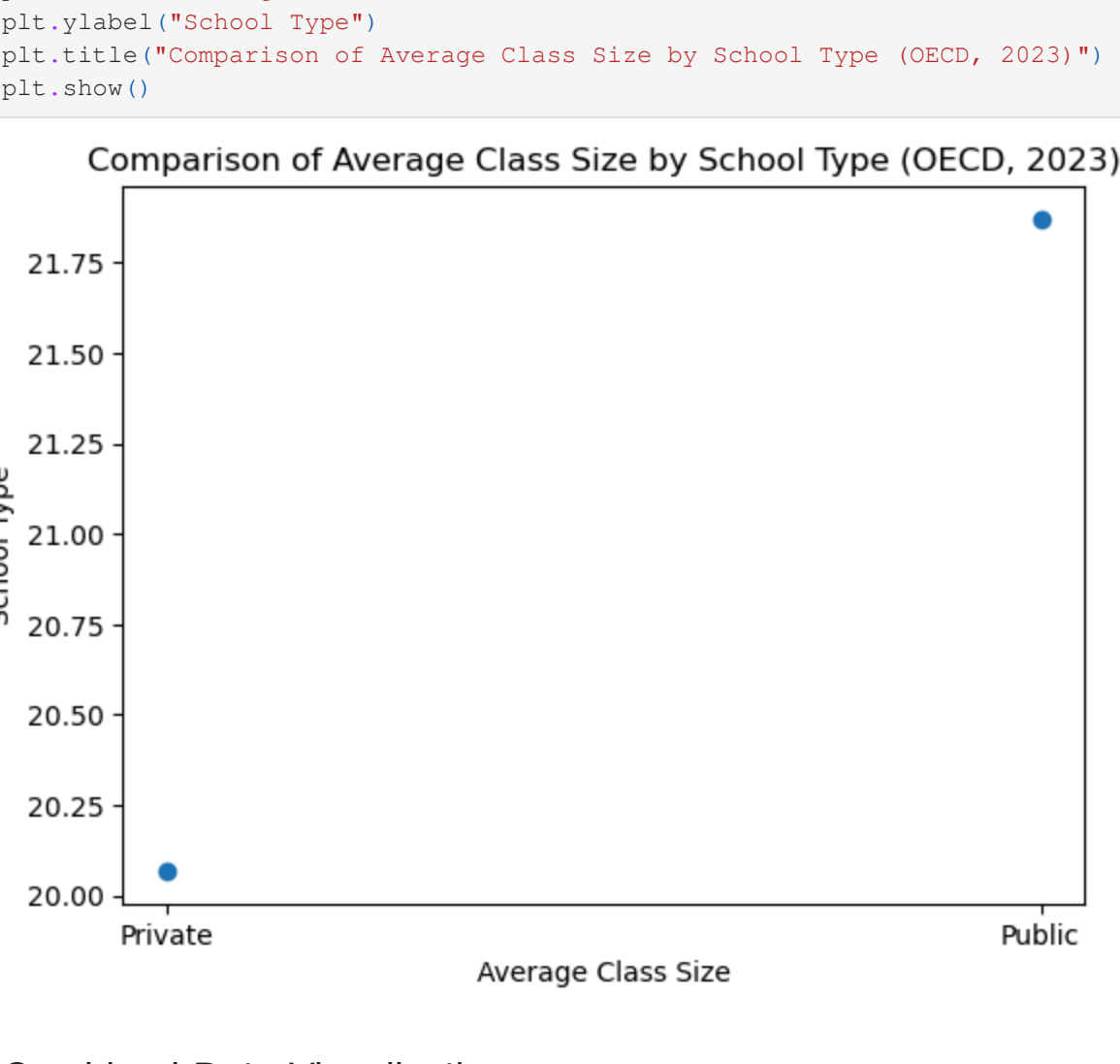
- The first visualization presents a bar chart comparing the average class size of public and private educational institutions using OECD data from 2023. The results show that public institutions have a larger average class size (approximately 21.9 students per class) than private institutions (approximately 20.1 students per class). This suggests that, on average, classrooms in public schools tend to include more students than those in private schools across OECD countries.
- The second visualization displays the same information using a dot plot, which emphasizes the magnitude of the difference between school types. Although the difference in average class size is modest, the separation between public and private institutions is clearly visible. Presenting the data in two complementary formats reinforces the observed pattern and confirms that public schools consistently operate with slightly larger class sizes. These extracted visualizations provide important contextual information that can help interpret differences in student performance observed in subsequent analyses.

```
In [47]: import matplotlib.pyplot as plt

plt.figure()
plt.bar(
    tidy_data['School_Type'],
    tidy_data['Average class size']
)
plt.xlabel('School Type')
plt.ylabel('Average Class Size')
plt.title('Average Class Size by School Type (OECD, 2023)')
plt.show()
```



```
In [48]: plt.figure()
plt.scatter(
    tidy_data['School_Type'],
    tidy_data['Average class size']
)
plt.xlabel('Average Class Size')
plt.ylabel('School Type')
plt.title('Comparison of Average Class Size by School Type (OECD, 2023)')
plt.show()
```



Combined Data Visualization

- The first combined visualization presents boxplots of exam score distributions by school type, using student-level data from Project 1. The plot shows that both public and private schools have similar median exam scores, with substantial overlap in their distributions. However, the public school distribution exhibits a slightly wider spread and more high-end outliers, indicating greater variability in exam performance among public school students.
- The second visualization integrates the Project 1 exam score data with the OECD API data on average class size. This bar chart compares the average exam score and average class size for public and private schools. While average exam scores are similar across school types, public schools are associated with a larger average class size. This comparison suggests that differences in classroom structure may coexist with relatively similar average performance outcomes. Although this analysis does not establish causality, it highlights how external contextual factors such as class size can provide additional insight when interpreting patterns in student performance.

```
In [49]: merge_df = pd.merge(df, tidy_data, how='left', on = 'School_Type')
merge_df.head()
```

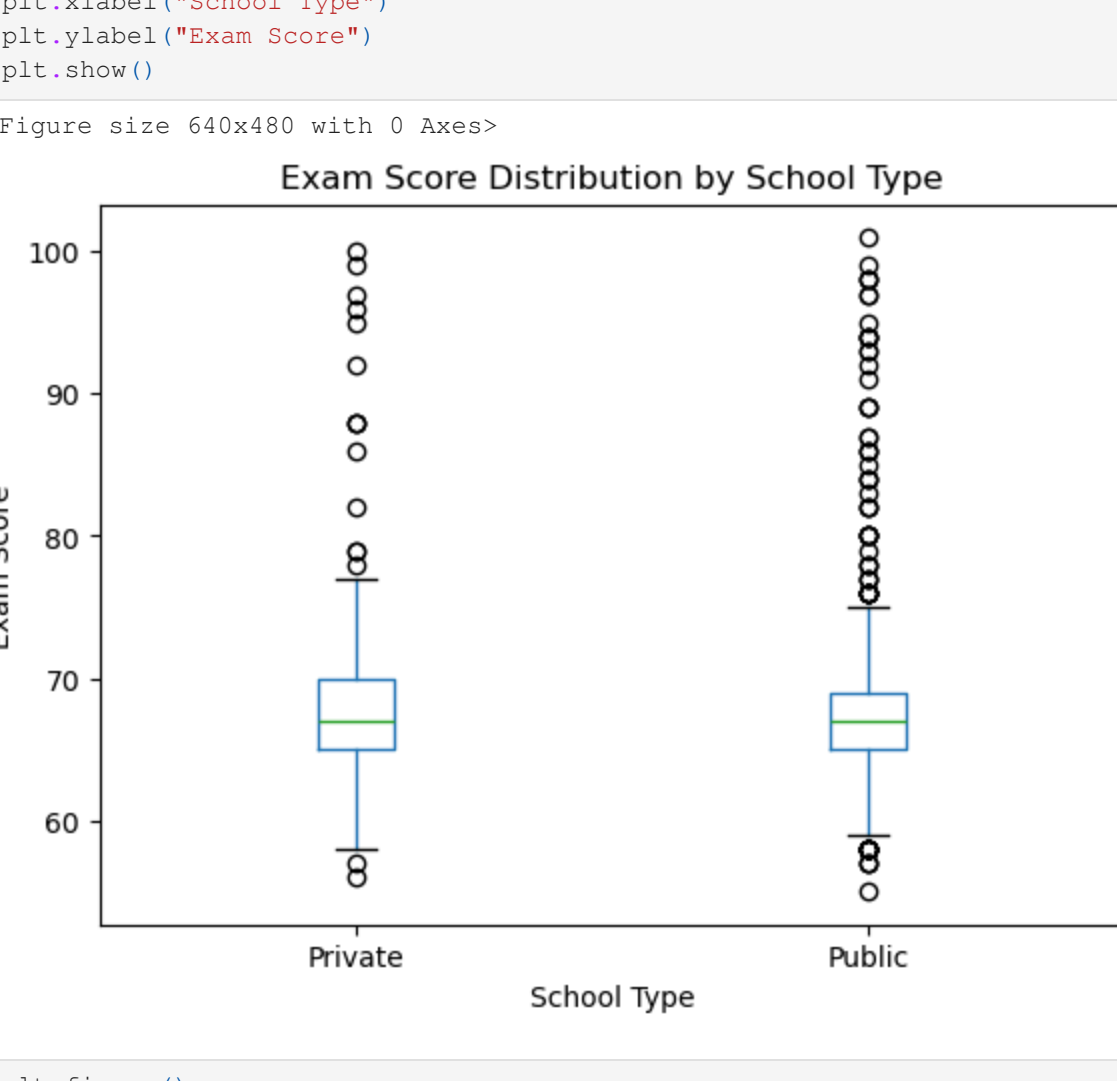
	Hours_Studied	Attendance	Parental_Involvement	Access_to_Resources	Extracurricular_Activities	Sleep_Hours	Previous_Scores	Motivation_Level	Internet_Access	Tutoring_Sessions	...	School_Type	Peer_Influence	Physical_Activity	Learning_Disabilities	Parental_Education_Level	Distar
0	23	84	Low	High	No	7	73	Low	Yes	0	...	Public	Positive	3	No	High School	
1	19	64	Low	Medium	No	8	59	Low	Yes	2	...	Public	Negative	4	No	College	
2	24	98	Medium	Medium	Yes	7	91	Medium	Yes	2	...	Public	Neutral	4	No	Postgraduate	
3	29	89	Low	Medium	Yes	8	98	Medium	Yes	1	...	Public	Negative	4	No	High School	
4	19	92	Medium	Medium	Yes	6	65	Medium	Yes	3	...	Public	Neutral	4	No	College	

5 rows × 22 columns

```
In [50]: plt.figure()
merge_df.boxplot(
    column='Exam_Score',
    by='School_Type',
    grid=False
)

plt.title('Exam Score Distribution by School Type')
plt.suptitle('') # removes automatic pandas title
plt.xlabel('School Type')
plt.ylabel('Exam Score')
plt.show()

<Figure size 640x480 with 0 Axes>
```

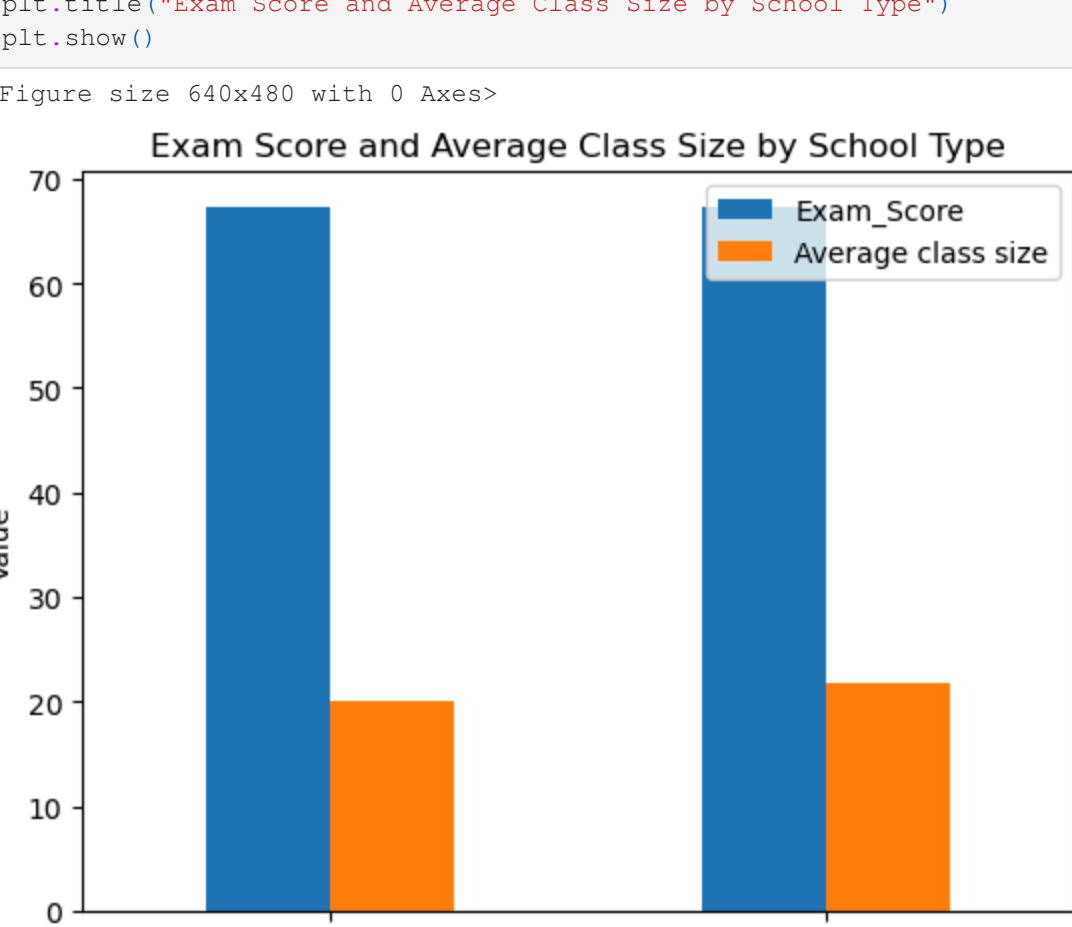


```
In [51]: plt.figure()

(
    merge_df
    .groupby('School_Type')[['Exam_Score', 'Average class size']]
    .mean()
    .plot(kind='bar')
)

plt.xlabel('School Type')
plt.ylabel('Value')
plt.title('Exam Score and Average Class Size by School Type')
plt.show()

<Figure size 640x480 with 0 Axes>
```



Next Steps and Hypothesis

In Project 1, I hypothesized that student performance may be influenced by factors beyond study time, such as school type. Additionally, some omitted factors correlated with school type may also affect exam outcomes, one of which is average class size. The combined analysis largely supports this expectation, although differences in average exam scores between school types are modest and their distributions overlap substantially. Incorporating OECD data on average class size provides important context for these results. Public schools are associated with larger average class sizes than private schools, which may contribute to greater variability in student performance observed in public institutions. However, the similarity in average exam scores suggests that class size alone does not fully explain performance differences. Future analyses could examine how class size interacts with instructional quality, resources, or student demographics. Overall, integrating external data strengthens the original hypothesis by placing student performance within broader educational structures.